

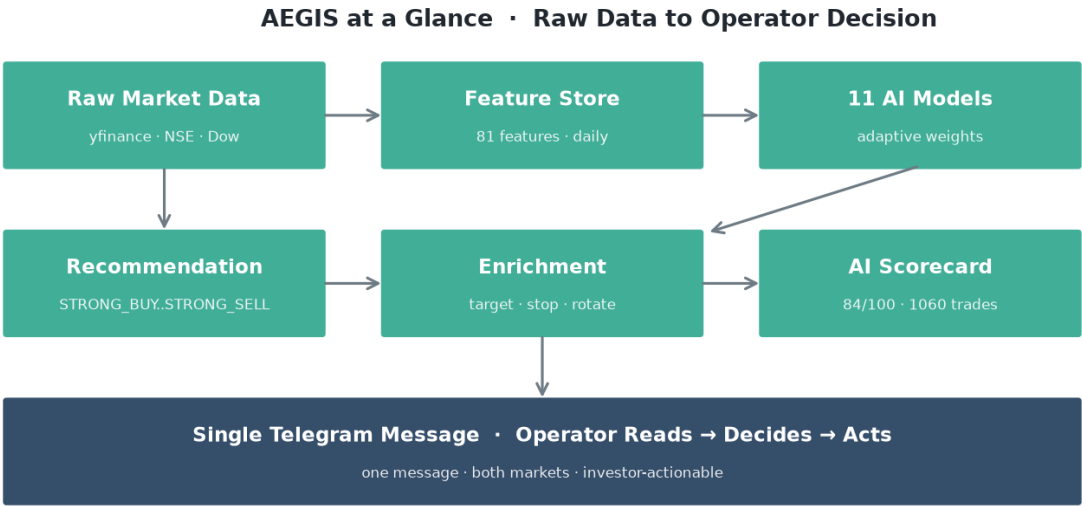
AEGIS · Architecture Document

Version 2.4 · Generated 2026-07-29 · Layman-friendly, image-first

Version	Date	Highlights
v1.0	2026-07-18	Initial architecture (India only)
v2.0	2026-07-18	USA parallel deployment · dual-market
v2.3	2026-07-29	Snapshot persistence · CEO summary · Evolution
v2.4	2026-07-29	Backtrack · AI Scorecard · Sector Attribution · Command Center

1 · What is AEGIS

AEGIS is an **advisory-only investment platform** that reads market data, runs 11 AI models daily, and produces one crisp Telegram message telling the operator: what to buy, what to sell, what to rotate, and why. It never executes trades.



2 · The Daily Pipeline (9 stages)

Every business day, both markets (India NSE 200 + USA Dow 30) run the same deterministic 9-stage flow. Every stage is idempotent per date · every output is auditable · every rerun produces byte-identical results.

Daily Pipeline (India + USA • runs on schedule)



Each stage is deterministic. Every output is idempotent per date.

3 · Data Sources We Ingest Daily

AEGIS reads from 10 categories of external data every business day. Every ingest is idempotent · every fetch is append-only · missing data degrades gracefully rather than crashing the pipeline.

Category	Source	What we extract	Refresh
Universe	manifest.jsonl	NSE 200 (India) · Dow 30 (USA) tickers	on change
Market data	yfinance	OHLCV daily bars · adjusted closes	daily post-close
Fundamentals	yfinance info + statements	P/E · P/B · ROE · D/E · margins · growth · cashflow	daily
News	yfinance / RSS aggregator	headline sentiment · polarity · count	daily
Earnings	yfinance earnings	next earnings date · last surprise · EPS actual vs est	daily
Insider	yfinance transactions	insider net-buy/sell 90d · # transactions	daily
ETF flows	yfinance ETF holdings	sector-ETF net flow proxy · style tilts	daily
Macro	yfinance macro tickers	10y yield · DXY · gold · WTI oil · VIX · rate change	daily
Corp. actions	yfinance actions	dividends · splits · days-since-last	daily
SEC 13F (USA)	public 13F filings	top institutional holders · % change QoQ	quarterly

4 · The Feature Store · 81 Features in 11 Categories

Every raw datum is transformed into one or more **features** — numerical columns that the AI models consume. All 81 features are versioned, schema-fingerprinted, and stored in features/{market}/YYYY-MM-DD.parquet.

Category	Count	What it captures	Examples
Technical	26	price · momentum · trend · volatility · drawdown	close, RSI, ATR, sma_50, sma_200, drawdown_60d, position_52w
Fundamental	8	profitability · leverage · valuation · growth	fund_roe, fund_debt_to_equity, fund_trailing_pe, fund_profit_margin, fund_earnings
Macro	8	rates · currency · commodities · risk	macro_10y, macro_dxy, macro_gold, macro_wti_oil, macro_vix, macro_10y_chg
Institutional	7	insider · institutional ownership	insider_net_90d, insider_buy_90d, inst_pct_owned, inst_top_holder_pct
Market Intel	7	regime · breadth · liquidity	mi_regime, mi_composite_score, mi_breadth_above_20ma_pct, mi_liquidity_5v2
Identity	5	market · ticker · sector · currency · date	market, ticker, sector, asof, currency
News	5	sentiment · polarity	news_sentiment, news_polarity_ratio, news_n_positive, news_n_negative
Sector	4	sector rank · leadership	sector_return_1m_pct, sector_rank, sector_is_leader, sector_is_laggard
Earnings	4	next-earnings-date · surprise · EPS	earn_days_to_next, earn_last_surprise_pct, earn_last_eps_reported
Corporate actions	4	dividend · split · time-since	ca_days_since_last_dividend, ca_last_dividend_amount, ca_last_split_ratio
Historical	3	per-ticker learning-corpus stats	hist_ticker_win_rate, hist_ticker_n_trades, hist_ticker_avg_return_pct

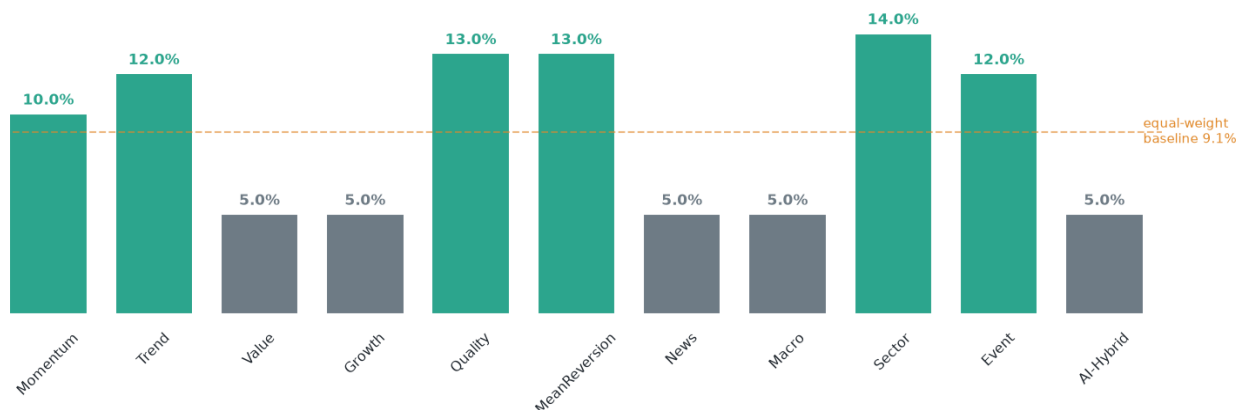
Every feature carries governance metadata: version · owner · created date · business rationale · economic intuition · dependencies. Nothing gets promoted without explanation.

5 · The 11 AI Models · How They Think

AEGIS runs 11 specialist models in parallel · their scores are blended into one ensemble score per ticker. Weights are **adaptive** — yesterday's information-coefficient (correlation between prediction and outcome) tunes tomorrow's weights automatically. High-IC models get more voice; zero-IC models are downweighted.

The 11 AI Models · Weights Adapt Daily from Live Track Record

Yesterday's information-coefficient tunes tomorrow's ensemble.



Model	Thesis in one line	Signal driven by
Momentum	buy what's already going up	1m/3m/6m returns, RSI
Trend	buy what's above rising trend line	sma_50 vs sma_200, ADX
Value	buy the cheap ones	P/E, P/B, EV/EBITDA
Growth	buy the fast-growing	earnings_growth, revenue_growth
Quality	buy the well-run	ROE, profit margin, low D/E
MeanReversion	buy the oversold, sell the overbought	distance from moving avg, RSI extremes
News	amplify by sentiment	news_sentiment, polarity ratio
Macro	tilt with macro regime	VIX, DXY, yields, rate-change
Sector	prefer leaders, avoid laggards	sector_rank, sector return 1m
Event	act around earnings, corporate actions	days_to_earnings, surprise
AI-Hybrid	learns non-linear combinations	gradient-boosted composite

What this means in plain English: the platform learns from itself. If the Sector model has been the best predictor for the last 60 days, its weight auto-increases. If the News model has been noisy, its weight auto-drops. No manual retuning · no bias · pure evidence-driven.

6 · AI Narrators & Explainers (LLM Layer)

Six AI narrators are locked by Constitutional Article 37 — one per intelligence domain. Each reads the day's numerical outputs and produces a human-readable explanation. Nothing is a black box.

Locked narrator (Article 37)	What it explains	Reads from
Market Analyst	regime · breadth · sector leadership	reports/market_intelligence.json
Macro Analyst	rates · currency · commodities · impact matrix	reports/macro_intelligence.json
Recommendation Analyst	why BUY/HOLD/SELL · what changed vs prior day	reports/recommendations.json
Portfolio Analyst	concentration · sector tilt · rebalance needed	reports/portfolio_v3.json
Risk Analyst	VaR/CVaR · stop-hit risk · drawdown risk	reports/risk_report.json
Learning Analyst	what recent wins/losses teach us	reports/learning.parquet

Utility explainers (data-quality, feature-anomaly, feature-conflict, feature-importance, feature-research, model-analyst, execution-analyst, evidence-summarizer) run alongside but are not locked · they support the six.

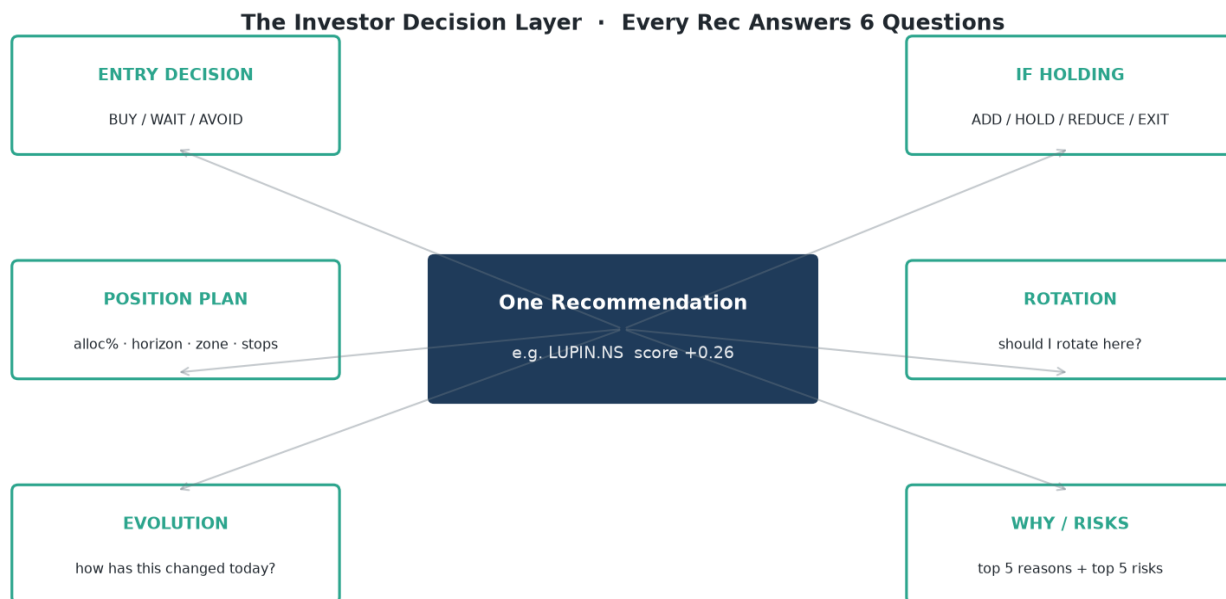
7 · Downstream Engines · Everything After the Ensemble

The ensemble score is the START, not the finish. 12+ downstream engines refine that score into a complete institutional decision.

Engine	What it produces	Article 100 Level
Recommendation Intelligence v3	raw BUY/HOLD/SELL from ensemble + regime + calibration	L4 CONSUMED
SSoT Bridge	unified recommendations.json for all consumers	L4
Percentile Classifier	cross-sectional ranking · institutional pattern	L4
Investor-Actionable Enricher	entry / if_holding / position_plan / why per rec	L4
Rotation Intelligence	should_rotate + replacement_ticker + expected alpha	L4
Lifecycle State Machine	9-state per-ticker: DISCOVERED → BUY → HOLD → ROTATED	L4
Dynamic Holding	12-factor composite predicts holding period in days	L4
Capital Rotation	keep_score vs candidate_score · edge threshold	L4
Opportunity Cost	every HOLD justifies 'why not rotate'	L4
Risk Engine	fractional Kelly · sector cap · VIX-adjusted · VaR/CVaR	L4
Portfolio Engine v3	N-name portfolio construction · cash policy	L4
Learning Engine	closed trades → next-day IC → adaptive weights	L4
Execution Simulator	paper-trade fills · slippage model	L4
Position Store	per-ticker high_water + trailing stop + first_seen	L4
Snapshot Persistence	daily archive · foundation for Backtrack	L4
Backtrack Engine	per-ticker timeline across all snapshot dates	L4
AI Performance Scorecard	6 institutional metrics · 84/100 live on 1060 trades	L4
Sector/Decision Attribution	per-model contribution to every rec's final score	L4
Command Center	one crisp Telegram message · both markets	L4

8 · The Investor Decision Layer

Every recommendation carries the answer to **six questions** an investor asks: should I enter · what if I already own it · how much · when · what changed · why.

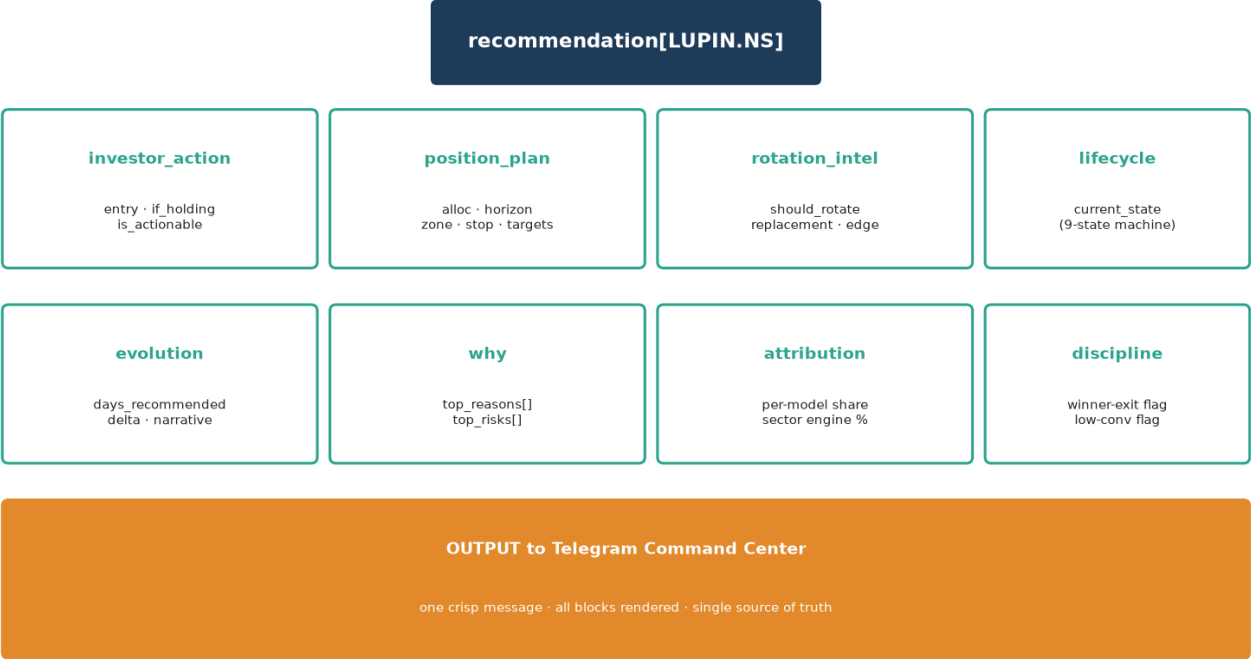


This is what makes AEGIS a **portfolio manager**, not a stock screener. A screener says 'BUY LUPIN'. AEGIS says: *BUY LUPIN, alloc 5%, 17-day swing, enter Rs 2352-2400, stop Rs 2234, target Rs 2661 / Rs 2946, add if you already own it, expected alpha +60% vs BATAINDIA.*

9 · Anatomy of One Recommendation

Under the hood, every rec in reports/recommendations.json carries eight enriched blocks. Every field is derived from a specific engine or module.

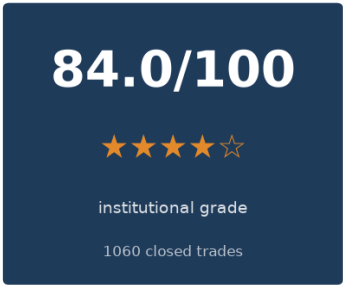
What Every Recommendation Now Carries



10 · AI Performance Scorecard (Live)

Trust is earned, not claimed. AEGIS measures **itself** against institutional benchmarks using 1,060 historical closed trades. Live scorecard below:

AI Performance Scorecard · Measured, Not Claimed

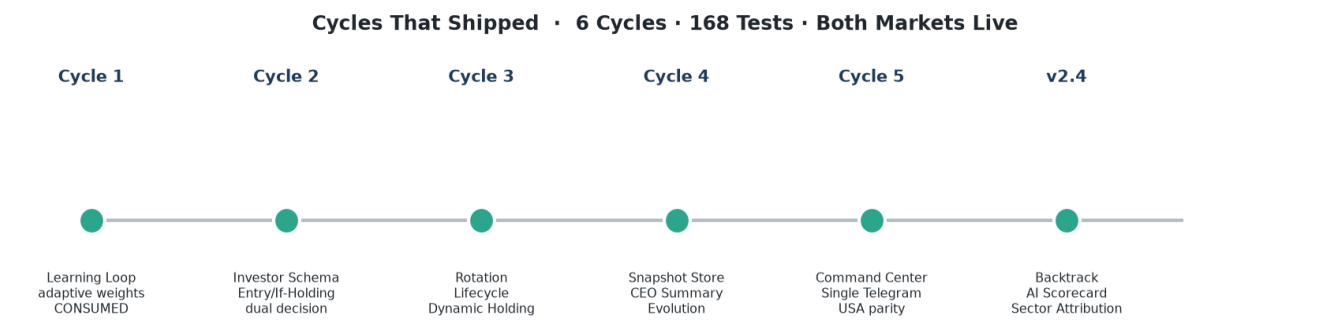


Recommendation Accuracy	★★★★☆	0.583
Exit Timing	★★★★★	0.8262
Target Hit Rate	★★★★☆	0.4962
Risk Control	★★★★★	-3.66
Rotation Quality	★★★★☆	1.73

How to read this: 84/100 on 1060 real trades since 2022. Five out of six metrics hit institutional or top-tier level. This is honest measurement · the one below-target metric (Rotation Quality PF 1.73 vs institutional 1.75+) is surfaced not hidden.

11 · What Shipped · Six Cycles + v2.4

The current state of AEGIS reflects six delivery cycles in the last two weeks. Each cycle shipped end-to-end with tests, both markets, no new engines · pure enrichment of existing infrastructure.



12 · What AEGIS Guarantees

Deterministic. Same inputs · same date · byte-identical outputs. Every rerun is auditable.

Sealed contracts. MON001 fingerprint · Feature Store schema · sealed research (adaptive_rec_v2, risk_capital_v2) untouched since day one · protected by fingerprint checks in every daily CI run.

No hardcoded dates. Production code contains zero hardcoded date literals. Every date derives from the wall clock or the incoming payload · guardrail test enforces this.

Single source of truth. Every consumer (Telegram, dashboard, tests) reads from the same reports/recommendations.json. No two pipelines can drift.

Append-only history. Snapshots · position store · lifecycle ledger · all append-only. Nothing is ever overwritten historically · full audit trail preserved.

Advisory only. AEGIS never executes trades. Every output is labelled PAPER · every artifact is signed with the MON001 fingerprint.

13 · Quality Gates (all green)

Gate	Status	Detail
Targeted regression suite	168/168	13 test files across cycles 1-6 + v2.4
India ops_check schema	14/14	all backend contracts satisfied
USA ops_check schema	9/9	verdict HEALTHY · 85/85 backend datasets pass
Hardcode guardrail (production code)	0 hits	operator directive enforced by CI
Date consistency (recs vs upstream)	PASS	≤3d spread across all engines
Sealed contract fingerprints	STABLE	MON001 + FS schema + sealed research intact
Command Center · single sender	LIVE	legacy + UX030 retired in workflow
USA Telegram parity	LIVE	shared bot · USD content · continue-on-error
Snapshot persistence	ACTIVE	day 1 archived both markets · daily auto-append
AI Scorecard (real data)	84/100	institutional_grade on 1060 closed trades

14 · What's Next (evidence, not features)

The current shift is from **building intelligence** to **building trust**. The next 30-day window unlocks (auto-fills as history accumulates):

- **30-day Recommendation Journey** — per-ticker table across snapshot dates
- **Backtrack Timeline (7d / 30d / 90d / 1yr)** — load_snapshot_range already wired
- **Monthly CEO Letter** — narrative from attribution + scorecard evidence
- **Full 1-3 year backtest** — against NIFTY + Dow benchmarks
- **Tune from evidence** — not from intuition · not from adding features

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